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PI Name:	Dutta, Prabir Ph.D.		
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PI Email:	dutta.1@osu.edu	**Fax:**	FY 614-688-5402
PI Organization Type:	UNIVERSITY	**Phone:**	614-292-4532
Organization Name:	The Ohio State University		
PI Address 1:	Department of Chemistry		
PI Address 2:	100 West 18th Avenue		
PI Web Page:	http://www.chemistry.ohio-state.edu/		
City:	Columbus	**State:**	OH
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Contact Monitor:	Sibille, Laurent	**Contact Phone:**	256 544-5221
Contact Email:	laurent.sibille@msfc.nasa.gov		
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Task Description:	Conventional synthesis of microporous materials usually involves mixing together reactants in an aqueous medium and results in immediate formation of an amorphous material, usually referred to as the gel from which crystal growth occurs. The complexity of the reaction mixture makes it difficult to examine the process. In our synthesis procedure, reactants are dispersed in small droplets of water in an oil-like medium with the help of detergents. These nanoclusters of water are called reverse micelles. The reactants in two different micelles can exchange with one another upon collision and if the conditions are appropriate, nucleation can occur. Our focus has been on the synthesis of zincophosphates, referred to as ZnPO. The first microporous material, zincophosphate sodalite was made using AOT reverse micelles. Considerable efforts were made to make other framework structures by primarily changing the Zn and phosphate concentrations in the AOT reverse micelles, and were unsuccessful. Recent experiments with the cationic reverse micelles DODMAC have led to the formation of open framework ZnPO-X, the structural analogue of the important gasoline cracking catalysts (zeolite Y or topoplogy). We have now broadened the scope of this synthetic method by synthesizing ZnPO-X with TOMAC reverse micelles. Our primary interest is a fundamental understanding of the molecular assembly of microporous materials. Can that goal be attained by examining crystal dissolution? We believe that to be the case for several reasons. First, the most interesting microporous frameworks are metastable structures and evolve to more condensed topologies with time. Second, clues regarding the complexity of the dissolution fragments of different frameworks provide information about the stability of the building units. There are two aspects of microgravity that we want to exploit. These include lack of sedimentation and absence of buoyancy driven convection. We have demonstrated that, under particular conditions of reverse micelle composition, crystal growth occurs in a layer-by-layer fashion. Avoiding sedimentation in these systems will lead to the growth of larger crystals. Light scattering will be used to follow the growth of crystals in microgravity. Seeding experiments to grow larger crystals is also possible, something that has not been accomplished in conventional synthesis. The second aspect of the microgravity experiments will focus on microprobe Raman studies, and absorption/fluorescence microscopy based on pH sensitive dyes. By comparing results in 1g and microgravity, we will discover if different surface intermediates are present during dissolution of different frameworks. Microporous materials include a large group of solids of varying chemical composition as well as porosity. These materials are characterized by channels and cavities of molecular dimensions. The framework structure is made up of interconnecting T-O-T' bonds, where T and T' can be Si, Al, P, Ga, Fe, Co, Zn, B and a host of other elements. Materials with Si-O-Al bonding in the framework are called zeolites and are extensively used in many applications. Ion-exchange properties of these materials are exploited in the consumer and environmental industries. Chemical and petroleum industries use zeolites as catalysts in hydrocarbon transformations. Synthesis of new microporous frameworks has led to the development of new technologies, and thus considerable effort worldwide is expended in their discovery. The complexity of the synthesis process and limited understanding of it has made it difficult to develop directed synthesis of microporous materials and most advances in this field have been made by trial and error. We are addressing the following basic issues in crystal growth of these materials including - Nature of the nucleation process - Molecular structure and assembly of nuclei - Growth of nuclei into crystals - Morphology control - Transformation of frameworks into other structures. Microporous materials are crystalline frameworks with well-defined cages and channels of molecular dimensions varying from 2 to 16 Angstroms. Various elements such as silicon, aluminum, zinc, phosphorus, gallium and transition metals can form the backbone of the framework. The most commonly used microporous frameworks are aluminosilicate zeolites. These find extensive use in detergents and catalysts. The ion-exchange of the extra framework ions is exploited in detergent use. The catalytic behavior arises because superacidity can be introduced into these frameworks and the topography of the channels and cages influence product formation. Discovery of novel frameworks can have significant impact on practical applications, as in the recent introduction by Unilever of a new zeolite in detergents and the use of ZSM-5 in petroleum cracking. Thus, there is considerable effort worldwide in the synthesis of new frameworks. Our contributions in understanding the basic molecular events that are responsible for formation and dissolution of microporous materials should eventually lead to more directed synthesis of novel framework structures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study has provided basic information on the mechanism of growth of microporous materials and application of nanometer sized microporous crystals.
	This report summarizes our research on this project over the lifetime of the grant: I. Morphological Control and Crystal Dissolution Pathways: Morphological control of microporous crystalline materials via synthetic routes is important for several reasons. For instance, it provides information about parameters that control nucleation and crystal growth, thereby illuminating aspects of the synthesis mechanism. Also, control of morphology is important for targeting various applications. We have shown that by altering composition, the morphology ZnPO-X could be changed. Previously reported synthesis of ZnPO-X at 4oC led to the formation of crystals with the characteristic octahedral morphology of faujasitic structures. Sizes of these crystals are in the <10 micron range. We found that by increasing the TMAOH/H3PO4 and Zn2+/Na+ ratios, the morphology was drastically altered to produce large hexagonal platelet type crystals with diameters around 100 micron. Based on electron microscopy data, we concluded that the presence of twin planes was leading to crystals with the platelet-like morphology. The possibility of synthesizing reasonably large flat crystals of ZnPO-X has made it feasible to do detailed Raman microprobe spectroscopic experiments. In particular, we have focused on microprobe Raman spectroscopy and examined how dissolution of ZnPO-X crystals occurs as a function of different monovalent cations in the medium. The vibrational information obtained from Raman spectroscopy made it possible to analyze the structural changes at the molecular level. In the presence of H+, at a pH of 3, the vibrational bands due to ZnPO-X disappeared completely in 12 minutes. These were replaced by Raman bands of hopeite, Zn3(PO4)2, a condensed form of zincophosphate. Electron microscopic experiments showed that the ZnPO-X was gradually getting covered with a film of hopeite. If, instead of H+, Li+, or Cs+ were used, then the ZnPO-X gradually converted to framework structures of LiA(BW) and CsZnPO4,

Task Progress:

respectively. How H+, Li+ and Cs+ destabilize in the ZnPO-X structure was also manifested in the vibrational band of tetramethylammonium (TMA) ions trapped in the sodalite cages of ZnPO-X. Upon exchange with H+, Li+, and Cs+, prior to collapse of the ZnPO-X and loss of TMA from the Raman spectra, there is a significant broadening of the TMA band at 765 cm-1. The bandwidth changes from ~ 6 cm-1 for Na+ to 11,12 and 16 cm-1 for H+, Cs+ and Li+, respectively. Our interpretation is that, upon ion-exchange with these cations, there is a distortion of the ZnPO-X framework, which makes the framework more susceptible to hydrolysis.

II. Influence of Framework Topology on Crystal Dissolution: In addition to Raman spectroscopy, we are developing several other techniques using to study the crystal dissolution process. The UV-Visible spectroscopic technique involves use a pH sensitive dye. Dissolution of ZnPO-X and sodalite crystals were carried out using a citrate buffer of pH 3, and methodology for measuring the differences in the dissolution dynamics were developed. The powder diffraction data clearly indicates that in both ZnPO-X and ZnPO-S dissolution, hopeite [Zn2(PO4)3] is being formed as the product at pH~3. Electron microscopy shows that the morphology of the hopeite formed in both cases is quite distinct. In the ZnPO-X, the hopeite crystals are considerably smaller and form a dense coating on the surface of the ZnPO-X pellet. For ZnPO-S, the hopeite crystals are considerably larger, and do not completely cover the pellet surface. Elemental analysis of the solution shows the presence of both zinc and phosphorus indicating the presence of M+(aq), Zn2+(aq) and PO43-(aq). Formation of hopeite occurs by the following reaction:3Zn2+ + 2PO43- Zn3(PO4)2 As the zincophosphate dissolves, supersaturation that initiates the nucleation of hopeite occurs around the pellet surface. The formation of hopeite is promoted because of its low solubility, Ksp being reported on the order of 10-35. A colorimetric method that monitors the pH change around the dissolving crystals provides a direct estimate of hopeite formation. We have examined the dissolution of hopeite and find no change in pH around the dissolving crystal. The dissolution of the framework zincophosphate structures with Zn/P = 1:1, to form hopeite leads to the imbalance of Zn and phosphate that leads to the pH change. The increase in pH around the dissolving crystal can be readily explained as arising from reaction of PO43- with H+ to form HPO42-/H2PO4-. Thus, any changes in pH that we observe during ZnPO dissolution can be related to the excess phosphate resulting from formation of hopeite. The larger change in pH for the ZnPO-X system is consistent with the elemental analysis data and supports the model that hopeite growth is more pronounced on ZnPO-X compared to the ZnPO-S system. The simplest possibility for hopeite formation is that the frameworks are dissolving to form Zn and phosphate, which then exceed supersaturation to form hopeite. However, considering the differences in morphology and the dynamics of hopeite formation and that the overall levels of solubilized zinc and phosphate are comparable, as seen from the elemental analysis, we do not favor this mechanism. A more likely possibility is that the hopeite is nucleated on the surface of the dissolving crystal from the dissolution products prior to the species escaping into the solution. We propose the hypothesis that the nuclei of hopeite arise from intermediate dissolution products of the zincophosphate frameworks. Because of the differences in the framework structure and the access of protons to the framework, the dissolution units are structurally different, and result in formation of hopeite of distinct morphologies. Small crystals are expected to grow faster than larger crystals for diffusion-controlled processes. The pH studies indicate that the hopeite formation on ZnPO-X is more pronounced than ZnPO-S and could be a reflection of the smaller crystals and diffusion-controlled growth. Influence of buoyancy-driven convection on the dissolution process is expected to be different for frameworks of different topologies.

III. Synthesis of Microporous Materials from Reverse Micelles

(a) ZnPO-S from DODMAC micelles: We have accomplished the synthesis of zincophosphate sodalite from DODMAC reverse micelles demonstrating the generality of this method. Seeding experiments and the growth of the crystals by light scattering have been demonstrated. Our growth experiments in microgravity will focus on the zincophosphate X and sodalite.

(b) Surface Characteristics of Crystals grown from Reverse Micelles: We have discovered that the novel growth process using reverse micelles leads to extraordinary smooth crystal faces, especially when compared to the conventional hydrothermal synthesis. We have hypothesized that the coating of water on the crystal in the reverse micelle growth disperse the nutrients uniformly over the crystal surface. The detergent helps with keeping the water layer on the crystal intact. Such differences have implications in both the growth mechanism and the usefulness of the crystals for various applications. From the SEM images of ZnPO-X synthesized by hydrothermal (HS) and reverse micelle (RM) methods, surface smoothness was evident on the crystals made by the RM method. This is more evident in the AFM images of the crystals. Previous studies have concluded, in general that zeolitic open frameworks both grow and dissolve by a layer-by-layer mechanism. Thus, at any time during crystal growth, a growing crystal face is bombarded by nuclei that can attach to the crystal face and start growing. Because of the random nature of the collisions, there are multiple nucleation sites on the crystal surface at which growth occurs. Eventually the growing faces meet and complete a layer on a surface, though during this time, crystal growth could have begun on top of the layer. Such a growth pattern leads to multiple layers and we could readily distinguish ~ 20-25 layers on the ZnPO-X crystal surface. For aluminosilicate faujasites, a structure with similar framework as ZnPO-X, Anderson and coworkers noted triangular terraces with step heights of approximately 15Å, which is the dimension of a faujasite sheet, and the present studies are consistent with these observations. Thus, it appears that the ZnPO-X HS sample has features very similar to those reported for aluminosilicate zeolites, and suggest similar growth mechanisms. On the other hand, crystals grown by the RM method show distinct differences from the HS grown crystals. Besides the differences in the surface roughness due to the lack of terraces on the (111) face, there is another important difference. The few triangular terraces noticed on the RM crystals have their orientations in registry with the overall crystals, with the apex of the triangle pointing to the crystal edges and not rotated, as was noted earlier for the HS grown crystals. Nucleation in the RM system begins in tiny water droplets that are dispersed in an organic medium. Crystal growth proceeds by acquiring nutrients from the aqueous interior of the reverse micelles that collide with the crystal surface. Since the zincophosphate crystal is growing in an organic medium and the crystal surface is polar, it is likely that the growing crystal is covered by a film of water separated from the hydrocarbon by a layer of surfactant molecules. Such an arrangement will stabilize the polar crystal surface-hydrocarbon interface. Thus, when reverse micelles containing the nutrients collide with the crystal surface, its contents get distributed in the water layer surrounding the crystal. This water layer provides a means for the nutrients to re-equilibrate and provide building blocks for the growth of the crystal. The absence of terraces in the RM samples indicates that terrace nucleation is slower as compared to propagation of the layer across the surface. The growth in the RM system is fundamentally different because of the presence of the protective surface water layer, which equilibrates and minimizes the number of species involved in crystal growth. Professor Michael George of University of Alabama, Huntsville was a collaborator on the atomic force microscopy for examining the crystal surfaces. Dr. N. Ramachandran from NASA MSFC developed the modeling associated with the crystal growth processes. He set up a model using the population balance approach and we have modeled growth using this model.

(c) Using microwave radiation for high temperatures in reverse micelles: The surfaces of zeolite crystals grown using reverse micelles showed significant differences then those microporous crystals synthesized via conventional hydrothermal method. Besides, this advantage, the reverse micelle method provides a convenient system to monitor the effects of each synthesis parameter on the growth of zeolite nanocrystals. For understanding the role of high temperature on reverse micelle, the growth of gold nanoparticles which is a well studied system was taken as an example. The temperature was increased by microwave heating, which basically increases the local temperature in the water cores of the reverse micelles. This alteration of the growth conditions did in fact increase the size of the gold nanoparticles and appeared to also promote a higher degree of crystallinity. Both the size and crystallinity of the gold nanoparticles were determined using transmission electron microscopy (TEM). Based on this observation, further studies are in progress to alter the growth conditions for the synthesis of larger and more crystalline microporous zincophosphate using microwave irradiation.

IV. Solid State Synthesis of Zincophosphate Faujasite (ZnPO-X): Structure Directing Role of 1,4-diazabicyclo[2,2,2]octane (DABCO)-Phosphate Complex: This study involves the structure directing behavior of 1,4-diazabicyclo[2,2,2]octane (DABCO) in the synthesis of a faujasite-like zincophosphate framework (ZnPO-X). The nucleation of ZnPO-X occurs within a minute of mixing of the reactants. The hypothesis we examine here is that the presence of a DABCO-phosphate complex in the synthesis medium promotes the nucleation of ZnPO-X. Circumstantial evidence in support of this hypothesis was obtained from analysis of the solids formed from a combinatorial mixture of the reactants DABCO, phosphoric acid, zinc nitrate and sodium hydroxide. The synthesis systems that had the DABCO and the phosphate in the same solution tended to form ZnPO-X at a faster rate. From compositions similar to that used in synthesis of ZnPO-X, we were able to synthesize a DABCO-phosphate complex, whose structure was determined by single crystal X-ray diffraction study. Using the DABCO-phosphate complex, ZnPO-X was prepared by a solid-state synthesis, and is the first report of solid-state synthesis of an open framework structure.

V. Seeded Crystal Growth: Synthesis of nano zeolites have been carried out from clear solution by many researchers in order to understand the early events that lead to nucleation and subsequent zeolite growth. However, it has been found that the yields of zeolites from such compositions are quite low, indicating that significant amount of reactants are left behind in solution. Further, even midsynthesis addition of reactants could lead to secondary nucleation indicating that primary building units may be present. These two observations imply that the mother liquor recovered after the crystallization should contain the primary building units or structures that can readily evolve into these building units. In order to manifest these observations we examined the mother liquor left behind after the completion of the crystallization of the nanocrystalline sodalite, zeolite A and Y from clear solutions. The mother liquor solutions were lypholized and examined by STEM. For few regions in the mother liquor of LTA and FAU, the electron diffraction pattern clearly suggested crystalline regions of about 2-5nm.

Further, many researchers speculated that the primary building units are distinct for each zeolite framework. We reasoned that if the primary building units are different, then solutions containing these units should exhibit a memory effect. We investigated if these mother liquors recovered from the nanocrystalline zeolite synthesis had any memory of the zeolites and can it direct the same compositions to different frameworks?

Our synthesis experiments involved addition of small amounts of mother liquor from which crystal growth has occurred to different compositions, with an intention not to alter the ratio of various components in these compositions, but enabling it to direct these compositions to the specific zeolite. Composition A 15.19(TMA)2O: 0.63 Na2O:1.6Al2O3: 44.9SiO2: 805H2O could be directed to sodalite or zeolite A, when the corresponding mother liquor was added. Thus, demonstrating that the solutions recovered from synthesis of nanocrystalline zeolites manifest a memory effect. However, with FAU mother liquor, composition A did not lead to zeolite Y, whereas compositions B (13.68 (TMA)2O:~0.05Na2O:2.188Al2O3:~9.34SiO2:805H2O) and C(~6.48 (TMA)2O:~0.05Na2O:2.188Al2O3:~9.34SiO2:805H2O) did. This suggests that in order for the species in mother liquor to manifest a memory effect, the nutrient composition is important. An interpretation of this observation is that the nutrient solutions need to provide species that act as the glue to combine the primitive units present in the mother liquor to viable nuclei. This hypothesis is also supported by dynamic light scattering studies. The growth studies of SOD mother liquor when added to Sodalite composition exhibit two growth processes. One occurred at ~20 hours and the second at~40 hours. The second part of the growth curve is what is expected for sodalite growth. The first part of the growth curve is arising from the primitive units present in the mother liquor and takes about 20 hours for crystal growth immaterial of composition to which it has been added. This indicates that the units in the mother liquor that grow into crystals are partially preformed and takes ~20 hours to grow into viable nuclei. Thus, this study shows that the particles that are left behind in the nanocrystalline zeolite synthesis do possess a memory and can grow to the corresponding nuclei if provided with the right building units.

VI. Nanometer-sized zeolite synthesis: In order to understand the dynamics of zeolite formation, we have examined the formation of nanocrystalline zeolites. This study has impacted our photochemical studies in two areas:

(a) Oxygen Transport in Zeolite Y Measured by Quenching of Encapsulated Tris(bipyridyl)ruthenium : This study deals with emission quenching of zeolite encapsulated trisbipyridyl ruthenium (II) (Ru(bpy)32+) by oxygen. Oxygen saturated solutions of Ru(bpy)32+ typically show about 70% quenching (I0/I = 3.3), where I0 and I are the peak intensities of the emission in N2 and O2, respectively. However, an aqueous suspension of Ru(bpy)32+-zeolite Na-Y (Si/Al= 2.5) (abbreviated as Ru-Na-Y) showed no quenching at all. This observation motivated us to analyze how the transport of O2 is occurring in the zeolite. Upon exposure of solid Ru-Na-Y (99% of intrazeolitic water) to N2/O2 dry gases, quenching in oxygen was found to be 5% (I0/I = 1.07). Partial dehydration at room temperature with loss of 33% of the water molecules from the zeolite led to 66% (I0/I = 2.96) quenching. Dehydration of Ru-Na-Y at 250 ?C under vacuum overnight led to complete loss of intrazeolitic water and increased quenching to 90% (I0/I = 10.7). Nanocrystalline Ru(bpy)32+-zeolite Y upon vacuum dehydration lost 55% of the intrazeolitic water and showed 96% (I0/I = 25.3) quenching. The extent of quenching of Ru(bpy)32+in zeolites by O2 is by far the largest as compared to previously studied matrices, and is being attributed to confinement of O2 in the supercages, which leads to increase in number of collisions with Ru(bpy)32+ and enhanced quenching. However, these samples showed complete lack of sensitivity (I0/I = 1) to oxygen upon exposure to water saturated gas or dissolved gas. Dealumination of zeolite framework by treatment with (NH4)2SiF6 produced a framework of Si/Al = 9.5, and with SiCl4 a framework of Si/Al >100. With increasing dealumination, the extent of quenching by dissolved O2 increased.

(b)Zeolite Mediated Photochemical Charge Separation Using A Surface Entrapped Ruthenium-Polypyridyl Complex : Employing the strategy of quaternization of the 2,2´ N atoms of the conjugated bipyridine ligand, 1,4-bis[2-(4'-methyl-2,2'-bipyrid-4-yl)ethenyl]benzene (L), a polypyridyl complex of ruthenium (II) has been tethered

	on the surface of zeolite Y. Electrochemical and spectroscopic properties of the ligand suggest that upon visible photoexcitation of the MLCT band, the electron is localized on the conjugated ligand rather than the bipyridines. Electron transfer from the surface complex to bipyridinium ions (methylviologen) within the zeolite was observed. Visible light photolysis of the ruthenium-zeolite solid ion-exchanged with diquat and suspended in a propylviologen sulfonate solution led to permanent formation of the blue propylviologen sulfonate radical ion in solution. The model that is proposed involves intrazeolitic charge transfer to ion-exchanged diquat followed by interfacial (zeolite to solution) electron transfer to propylviologen sulfonate in solution. Because of the slow intramolecular back electron transfer reaction and the forward electron propagation via the ion-exchanged diquat, Ru(III) is formed. This Ru(III) complex formed on the zeolite is proposed to react rapidly with water in the presence of light, followed by reaction with the propylviologen sulfonate to form pyridones and regeneration of Ru(II), which then continues the photochemical process.
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